

CENG381 Stochastic Processes

Poisson Process — Practice 1 (Answer Key)

Q1 Solution

a) PMF, mean, and variance:

$$P(N(t) = n) = (\lambda t)^n * e^{(-\lambda t)} / n!, \quad n = 0, 1, 2, \dots$$

$$E[N(t)] = \lambda t$$

$$\text{Var}(N(t)) = \lambda t$$

(For a Poisson distribution, mean = variance = λt .)

b) Specific probabilities for $t=2, \lambda=3 \rightarrow \lambda t=6$:

$$\begin{aligned} P(N(2)=6) &= 6^6 * e^{(-6)} / 6! = 46656 * e^{(-6)} / 720 \\ &= 64.8 * e^{(-6)} \approx 64.8 * 0.002479 \approx 0.1606 \end{aligned}$$

$$\begin{aligned} P(N(2) \leq 1) &= P(N(2)=0) + P(N(2)=1) \\ &= e^{(-6)} + 6 * e^{(-6)} = 7 * e^{(-6)} \\ &\approx 7 * 0.002479 = 0.01735 \end{aligned}$$

c) First email after time t :

$$\{\text{first email after } t\} = \{N(t) = 0\}$$

$$\begin{aligned} P(\text{first email after } t=1/2) &= P(N(1/2)=0) \\ &= e^{(-\lambda t)} = e^{(-3/2)} = e^{(-1.5)} \approx 0.2231 \end{aligned}$$

Q2 Solution

a) Thinned Poisson (latte orders):

Thinning property: Poisson(λ) thinned by $p \rightarrow$ Poisson(λp)

Latte rate = $10 * 0.4 = 4$ per hour

$$\begin{aligned} P(3 \text{ lattes in } 1 \text{ hr}) &= 4^3 * e^{(-4)} / 3! = 64 * e^{(-4)} / 6 \\ &= 32/3 * e^{(-4)} \approx 10.667 * 0.01832 \approx 0.1954 \end{aligned}$$

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b) Superposition (combined arrivals):

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Superposition:      Poisson(lambda)      +      Poisson(mu)      ->
Poisson(lambda+mu)
Combined rate = 10 + 6 = 16 per hour
In 15 min = 1/4 hour: mean = 16 * (1/4) = 4 arrivals

P(at least 2 in 15 min) = 1 - P(N(1/4)=0) - P(N(1/4)=1)
                        = 1 - e^(-4) - 4*e^(-4)
                        = 1 - 5*e^(-4)
                        ~ 1 - 5*0.01832 = 1 - 0.09158 = 0.9084
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c) Conditional expectation using independent increments:

The Poisson process has independent increments: $N_{\text{shop}}(3) - N_{\text{shop}}(1)$ is independent of $N_{\text{shop}}(1)$.

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E[N_shop(3) | N_shop(1)=4]
= E[N_shop(1) | N_shop(1)=4] + E[N_shop(3)-N_shop(1) |
N_shop(1)=4]
= 4 + E[N_shop(3)-N_shop(1)] [by independence]
= 4 + lambda*(3-1)
= 4 + 10*2 = 4 + 20 = 24
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Q3 Solution

a) Derivation of the forward equation:

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P_j(t+Delta) = P(N(t+Delta)=j)
              = P(no arrival in (t,t+Delta)) * P(N(t)=j)
              + P(one arrival in (t,t+Delta)) * P(N(t)=j-1)
              + o(Delta)
              = (1 - lambda*Delta + o(Delta))*P_j(t)
              + (lambda*Delta + o(Delta))*P_{j-1}(t) + o(Delta)

P_j(t+Delta) - P_j(t) = -lambda*Delta*P_j(t)
                      + lambda*Delta*P_{j-1}(t) + o(Delta)
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Divide by Delta and take Delta $\rightarrow 0$:

$$P_j'(t) = -\lambda P_j(t) + \lambda P_{j-1}(t) \quad \checkmark$$

b) Verification that the Poisson PMF satisfies the forward equation:

Let $P_j(t) = (\lambda t)^j e^{-\lambda t} / j!$. Differentiate:

$$\begin{aligned} & d/dt [(\lambda t)^j * e^{-\lambda t} / j!] \\ &= [j * (\lambda t)^{j-1} * \lambda * e^{-\lambda t} + (\lambda t)^j * (-\lambda) * e^{-\lambda t}] / j! \\ &= \lambda * e^{-\lambda t} * [(\lambda t)^{j-1} / (j-1)! - (\lambda t)^j / j!] \\ &= \lambda * P_{j-1}(t) - \lambda * P_j(t) \\ &= -\lambda P_j(t) + \lambda P_{j-1}(t) \quad \checkmark \end{aligned}$$

c) Bus waiting time ($\lambda=2$, $t=3/4$ hour for 45 min):

$$\begin{aligned} P(\text{first bus after 45 min}) &= P(N(3/4) = 0) \\ &= e^{-\lambda * 3/4} = e^{-2 * 3/4} = e^{-3/2} \approx 0.2231 \\ E[\text{wait time}] &= 1/\lambda = 1/2 \text{ hour} = 30 \text{ minutes} \end{aligned}$$

The inter-arrival time X_i is Exponential($\lambda=2$). $P(X_i > 3/4) = P(N(3/4)=0) = e^{-3/2} \approx 22.3\%$. The expected wait is $1/\lambda = 30$ min regardless of how long you have already waited (memoryless property of the exponential).